	Course Name: Advanced Web Technology	EXPERIMENT NO. 9
	Course Code: 20CP314P	Branch: CSE
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Objective: Working with React hooks

1. Write react programs displaying the usage of each of the below given react hooks:

a. `useReducer`

UseReducer.jsx

```
import { useReducer } from "react";
import { ACTION_TYPES } from "../postActionTypes";
import { INITIAL_STATE, postReducer } from "../postReducer";

const Post = () => {
  const [state, dispatch] = useReducer(postReducer, INITIAL_STATE);

  const handleFetch = () => {
    dispatch({ type: ACTION_TYPES.FETCH_START });
    fetch("https://randomuser.me/api/")
      .then((res) => {
        return res.json();
      })
      .then((data) => {
        const firstName = data.results[0].name.first;
        const lastName = data.results[0].name.last;

        const greet = { message: "Hi " + firstName + " " + lastName };

        dispatch({ type: ACTION_TYPES.FETCH_SUCCESS, payload: greet });
      })
      .catch(() => {
        dispatch({ type: ACTION_TYPES.FETCH_ERROR });
      });
  };

  return (
    <div>
      <h1>useReducer</h1>
      <button onClick={handleFetch}>
        {state.loading ? "Wait..." : "Fetch the post"}
      </button>
      <p>{state.post?.message}</p>
      <span>{state.error && "Something went wrong!"}</span>
    </div>
  );
};

export default Post;
```

postreducer.js

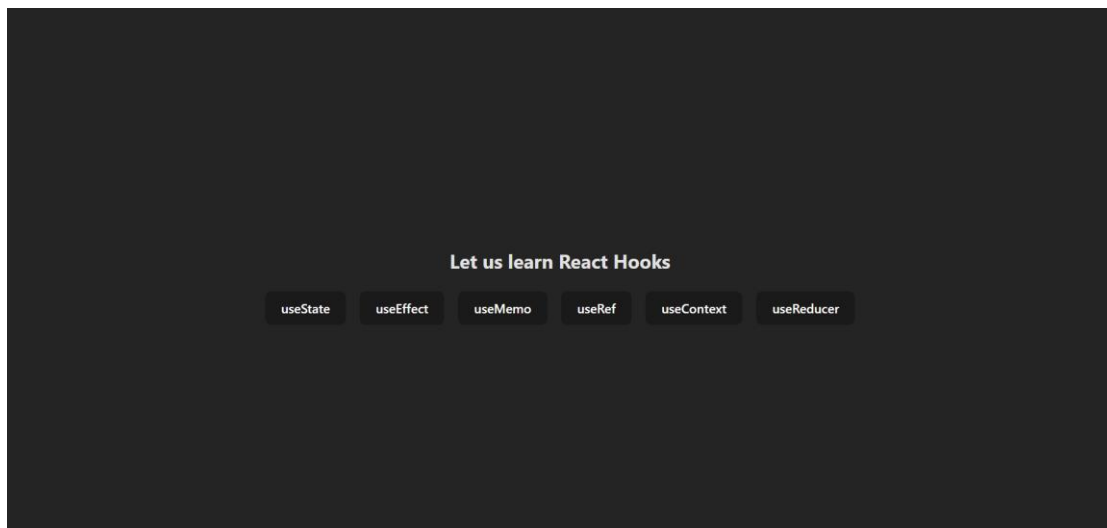
```
import { ACTION_TYPES } from "../postActionTypes";

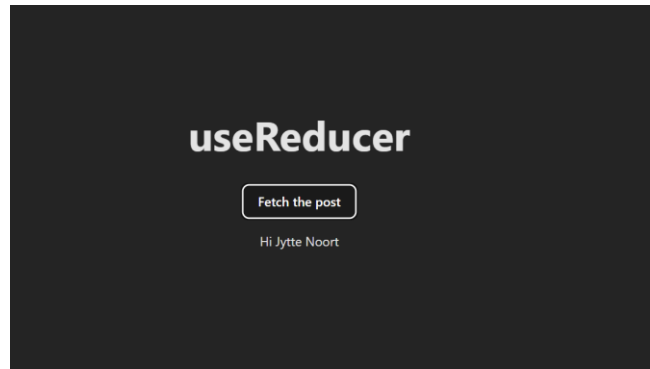
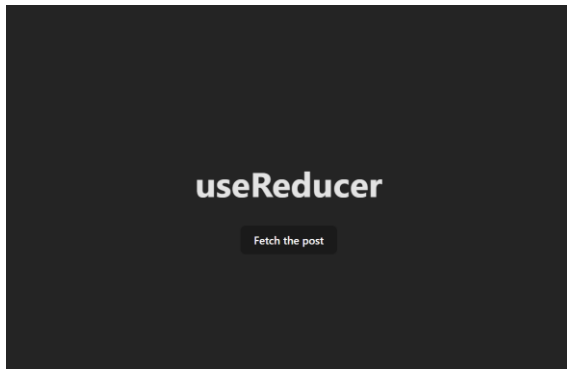
export const INITIAL_STATE = {
  loading: false,
  post: {},
  error: false,
};

export const postReducer = (state, action) => {
  switch (action.type) {
    case ACTION_TYPES.FETCH_START:
      return {
        loading: true,
        error: false,
        post: {},
      };
    case ACTION_TYPES.FETCH_SUCCESS:
      return {
        ...state,
        loading: false,
        post: action.payload,
      };
    case ACTION_TYPES.FETCH_ERROR:
      return {
        error: true,
        loading: false,
        post: {},
      };
    default:
      return state;
  }
};
```

postActionTypes.js

```
export const ACTION_TYPES = {
  FETCH_START: "FETCH_START",
  FETCH_SUCCESS: "FETCH_SUCCESS",
  FETCH_ERROR: "FETCH_ERROR",
};
```





b. useMemo

```
import { useState, useMemo } from "react";
import "../App.css";
function UseMemo() {
  const [count, setCount] = useState(0);
  const [darkTheme, setDarkTheme] = useState(false);
  const squareResult = useMemo(() => {
    delayFunction(count);
  }, [count]);
  const themeStyles = useMemo(() => {
    return {
      backgroundColor: darkTheme ? "blue" : "red",
      color: darkTheme ? "red" : "blue",
    };
  }, [darkTheme]);

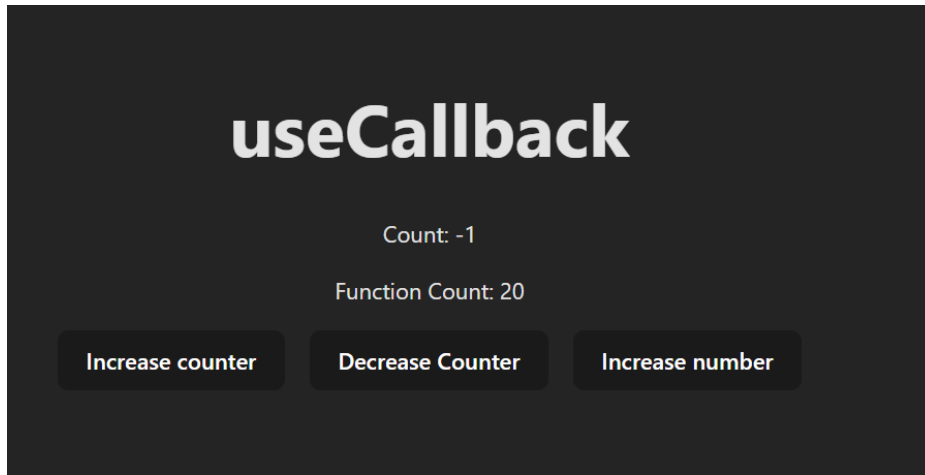
  return (
    <>
      <div className="card">
        <h1>useMemo</h1>
        <label>
          Number:{" "}
          <input
            type="number"
            value={count}
            onChange={(e) => setCount(parseInt(e.target.value))}
          />
        </label>

        <p>Square: {squareResult}</p>

        <button onClick={() => setDarkTheme((prevTheme) => !prevTheme)}>
          Change Theme
        </button>
        <p style={themeStyles}>Theme div</p>
      </div>
    </>
  );
}

function delayFunction(n) {
  for (let i = 0; i < 1000000000; i++);
  return n * n;
}
export default UseMemo;
```





d. useContext

```
import { useState, createContext, useContext } from "react";
```

```
const UserContext = createContext();
```

```
function useContext() {
  const [user] = useState("AWT Student");
```

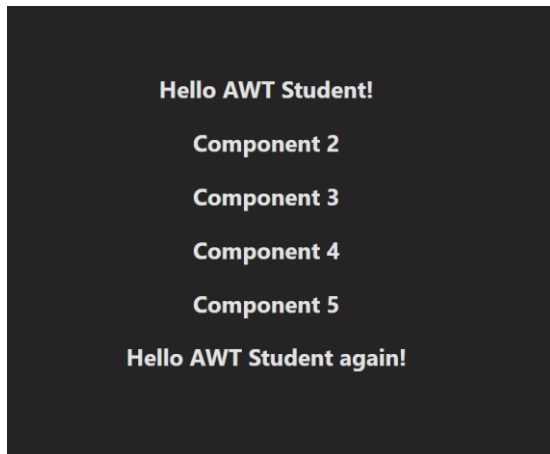
```
  return (
    <UserContext.Provider value={user}>
      <h2>{`Hello ${user}!`} </h2>
      <Component2 />
    </UserContext.Provider>
  );
}
```

```
function Component2() {
  return (
    <>
      <h2>Component 2</h2>
      <Component3 />
    </>
  );
}
```

```
function Component3() {
  return (
    <>
      <h2>Component 3</h2>
      <Component4 />
    </>
  );
}
```

```
function Component4() {
  return (
    <>
      <h2>Component 4</h2>
      <Component5 />
    </>
  );
}
```

```
function Component5() {
  const user = useContext(UserContext);
  return (
    <>
      <h2>Component 5</h2>
      <h2>`Hello ${user} again!`</h2>
    </>
  );
}
export default UseContext;
```



e. useEffect

```
import { useEffect, useState } from "react";
import "../App.css";
function UseEffect() {
  const [count, setCount] = useState(0);
  const [numberProperty, setNumberProperty] = useState("");

  useEffect(() => {
    if (count % 2 === 0) {
      setNumberProperty(`Even Number: ${count}`);
    } else {
      setNumberProperty(`Odd Number: ${count}`);
    }
    console.log("useEffect changed");
  }, [count]);

  return (
    <>
      <div className="card">
        <h1>useEffect</h1>
        <button onClick={() => setCount((count) => count + 1)}>
          Click Me!
        </button>
        <p>{numberProperty}</p>
      </div>
    </>
  );
}
export default UseEffect;
```



2. Design a custom hook that helps you declare a default value of a component

Customhook.jsx

```
import useFetch from "./useFetch";
const Customhook = () => {
  const data = useFetch("https://jsonplaceholder.typicode.com/todos");
  return (
    <div className="card">
      <h1>Custom Hook</h1>
      {data &&
        data.map((item) => {
          return <p key={item.id}>{item.title}</p>;
        })
      }
    </div>
  );
};
export default Customhook;
```

useFetch.js

```
import { useState, useEffect } from "react";
const useFetch = (url) => {
  const [data, setData] = useState(null);

  useEffect(() => {
    fetch(url)
      .then((res) => res.json())
      .then((data) => setData(data));
  }, [url]);

  return data;
};
export default useFetch;
```

